USAMA ALI

AI Engineer • ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience building AI applications using Python, RAG pipelines, LLM APIs, embeddings, and machine learning. Experienced in FastAPI backend development, PostgreSQL, semantic search, and integrating AI technologies into practical applications.

TOOLS & TECHNOLOGY

- **Programming Languages:** Python
- **AI & Machine Learning:** RAG, LLM APIs, Embeddings, LangChain, PyTorch, TensorFlow
- **Backend & Security:** FastAPI, REST APIs, SQLAlchemy, Alembic, JWT Authentication, Authorization
- **Databases:** PostgreSQL, SQLite
- **Vector Databases:** pgvector, ChromaDB, FAISS
- **Tools & Automation:** Pandas, NumPy, Git, GitHub API, n8n

EXPERIENCE

GenAI Engineer Intern — *CypherShield* July – November 2025

- Contributed to GenAI application workflows by integrating and working with LLM APIs.
- Built automation workflows using N8N to streamline recurring development tasks.
- Collaborated with team members to communicate requirements and support GenAI development activities.

SELECTED PROJECTS

RepoGuide — AI-Powered Codebase Q&A; (github.com/UsamaAliK/RepoGuide)

- Built a tool that analyzes GitHub repositories and provides context-aware answers about their codebase using RAG.
- Implemented repository ingestion, code-aware chunking, embeddings, semantic search, and reranking using Python, FastAPI, ChromaDB, Gemini, Jina AI, and LangChain.
- Implemented PostgreSQL persistence using SQLAlchemy and Alembic migrations for users, repositories, conversations, and messages.

Mindly — Knowledge-Based Learning System (mindly.one)

- Built a knowledge-based AI system using RAG, embeddings, and LLM APIs for context-aware responses.
- Implemented document ingestion and semantic retrieval using Python, Next.js, and pgvector.

Movie Recommendation System (github.com/UsamaAliK/Movie-Recomendation-System)

- Built a collaborative filtering movie recommendation system using Singular Value Decomposition (SVD).
- Implemented data preprocessing, model training, and recommendation generation using Python, Scikit-learn, and Surprise.

EDUCATION

Bachelor of Science in Computer Science 2022 – 2026

University of Management and Technology, Lahore